

Final Project (Extended from Min project)

Complete all the tasks till week 12

Marks: 15

Task 1: Convert the ERD of mini project (already completed by students) into a Relational Database

Using the ERD developed in Phase 1:

1. Convert all entities into relational tables.
2. Convert all 1:1, 1:M, and M:N relationships appropriately.
3. Create associative tables for M:N relationships.
4. Identify the primary key of every table.
5. Identify all foreign keys.
6. Define appropriate data types for every attribute.
7. Define NOT NULL, UNIQUE, CHECK, and DEFAULT constraints where appropriate.
8. Implement the recursive faculty mentoring relationship.
9. Implement relationship attributes such as:
 - supervision role and start date
 - student project role and contribution percentage
 - funding amount and funding date
 - publication author order and corresponding-author status
 - submission information
 - review scores and comments
 - revision information
 - equipment usage date and purpose

Deliverable: Complete SQL DDL script for creating the database.

Task 2: Database Normalization

Analyze the relational schema produced from the ERD.

Students should:

1. Identify possible functional dependencies.
2. Explain whether each table satisfies:
 - 1NF
 - 2NF
 - 3NF
3. Identify any possible insertion, deletion, and update anomalies.
4. Decompose any table that is not appropriately normalized.
5. Provide the final normalized schema.

Deliverable: A short normalization report with functional dependencies and the final schema.

Task 3: Populate the Database

Create realistic sample data.

Students should insert sufficient data to demonstrate the functionality of the system.

Minimum recommended data:

Table/Entity	Minimum Records
Departments	5
Programs	8
Students	30
Faculty	15
Research Projects	15

Table/Entity	Minimum Records
Funding Sources	8
Publications	25
Conferences	8
Conference Submissions	20
Reviews	30
Revisions	20
Equipment	10

Students should ensure that the data satisfies all primary-key, foreign-key, and business-rule constraints.

Deliverable: SQL data insertion script.

Task 4: Advanced SQL Queries

Develop SQL queries that demonstrate advanced database skills.

Students must write queries for:

1. Students and their academic programs.
2. Students and their supervisors.
3. Faculty supervising more than five students.
4. Projects involving more than three students.
5. Projects with total funding above a specified amount.
6. Students participating in multiple research projects.
7. Faculty members who are authors of publications.
8. Publications with more than three authors.
9. Publications where a particular faculty member is the corresponding author.
10. Conferences receive more than five submissions.

11. Submissions that received a particular decision.
12. Students who have never participated in a research project.
13. Faculty members who do not supervise any student.
14. Projects that use specific equipment.
15. Faculty members who mentor other faculty members.
16. Faculty members who have no mentor.
17. The most-funded research project.
18. The most productive author.
19. The department with the highest number of research projects.
20. The average contribution percentage of students in each project.

Require students to demonstrate:

- joins
- subqueries
- correlated subqueries
- aggregation
- GROUP BY
- HAVING
- EXISTS
- set operations where appropriate
- views

Task 5: Create Database Views

Students should create at least **five views** useful for university management.

For example:

1. StudentResearchProfile
2. FacultyResearchProfile

3. ProjectFundingSummary
4. PublicationAuthorSummary
5. ConferenceSubmissionStatus

For each view, students should explain:

- Why the view is required.
- Who should have access to it.
- What information it hides.
- Whether the view should be updateable.

Deliverable: SQL scripts and a short explanation of each view.

Task 6: Stored Procedures and Functions

Create at least:

Three stored procedures

Examples:

- Add a new research project.
- Register a student in a research project.
- Submit a publication to a conference.

Three functions

Examples:

- Calculate total project funding.
- Calculate the number of publications by a faculty member.
- Calculate the total contribution of students to a project.

Students should demonstrate parameter validation and appropriate error handling.

Task 7: Triggers and Business Rules

Implement database-level business rules using triggers.

For example:

1. Prevent a student's total project contribution from exceeding 100%.
2. Prevent a student from being assigned an invalid supervisor.
3. Prevent funding with a negative amount.
4. Automatically record changes to important project information.
5. Prevent deletion of a faculty member who is still assigned as a supervisor.
6. Automatically maintain an audit record when publication information is modified.

Students must explain **why the rule belongs at the database level** rather than only in the application.

Task 8: Database Security and Access Control: advanced database governance.

Create at least the following database roles:

Role 1: Database Administrator

Full database administration privileges.

Role 2: Research Administrator

Can manage:

- projects
- funding
- conferences
- publications

but should not have unrestricted access to sensitive student information.

Role 3: Faculty Researcher

Can:

- view relevant research projects
- manage their research information
- access permitted publication information

but cannot modify database structure.

Role 4: Student

Can:

- view their own profile
- view their projects
- view permitted publication/submission information

but cannot access other students' confidential information.

Role 5: Auditor

Can read audit records but cannot modify operational data.

Students should:

1. Create the roles.
2. Grant appropriate privileges.
3. Revoke inappropriate privileges.
4. Demonstrate least-privilege access.
5. Provide a privilege matrix.

Example:

Object	DBA	Research Admin	Faculty	Student	Auditor
Student	CRUD	R	Limited R	Own record	R
Faculty	CRUD	R	Own record	R	R
Project	CRUD	CRUD	R/limited C/U	R	R
Funding	CRUD	CRUD	R	No access	R
Publication	CRUD	CRUD	C/R/U	R	R
Audit Log	R/W	No	No	No	R

This task directly assesses **role-based access control and least privilege**.

Task 9: Data Governance

Develop a **Data Governance Plan** for the Research Management System.

They should define:

Data ownership

Who owns:

- student data?
- faculty data?
- research data?
- funding data?
- publication data?

Data stewardship

Identify the responsible data steward for each major data domain.

Data classification

Classify information into:

- Public
- Internal
- Confidential
- Highly Confidential

For example:

Data	Classification
Faculty name	Public/Internal
Student email	Confidential
Student academic status	Confidential
Research project title	Internal/Public
Funding amount	Confidential

Data	Classification
Reviewer comments	Highly Confidential
Password credentials	Highly Confidential

Data quality rules

Define rules for:

- completeness
- accuracy
- consistency
- uniqueness
- validity
- timeliness

Deliverable: 3 to 5-page Data Governance Plan.

Task 10: Database Auditing

Implement an auditing mechanism.

The system should record:

- user
- operation
- table
- record affected
- date/time
- old value
- new value

Students should demonstrate auditing for at least:

1. Student records.

2. Research projects.
3. Funding.
4. Publications.

Then answer:

Who should be allowed to view the audit log, and who should not?

Task 11: Data Privacy and Security

Identify sensitive information in the database.

1. Identify personally identifiable information.
2. Identify confidential research information.
3. Identify sensitive reviewer information.
4. Propose encryption requirements.
5. Propose password/security requirements if authentication data is included.
6. Explain which data should be masked.
7. Identify data that should not be visible to students.
8. Explain how unauthorized disclosure could occur.

Advanced requirement

Create a mechanism for **data masking** or restricted views so that a user does not see information they are not authorized to access.

Task 12: Backup and Recovery Strategy

Design a backup and recovery strategy for the database.

- Full backup frequency.
- Incremental/differential backup strategy.
- Backup retention period.
- Off-site backup.

- Backup encryption.
- Recovery procedures.
- Disaster recovery strategy.

Then define:

RPO: How much data loss is acceptable?

RTO: How quickly should the system be restored?

Example scenario:

The Research Management System database fails at 2:00 PM on Sunday because of a storage failure. The latest backup was completed at midnight.

Students should explain how they would recover the database and determine the potential data loss.

Task 13: Transaction Management and Concurrency

Create/simulate realistic concurrent transactions.

For example:

Two research administrators attempt to update the funding information of the same research project at the same time.

Students should investigate:

- transactions
- COMMIT
- ROLLBACK
- isolation levels
- dirty reads
- non-repeatable reads
- phantom reads
- lost updates
- locking

Students should demonstrate at least one concurrency problem and explain how it can be prevented.

Task 14: Performance Optimization

Students should analyze database performance.

Require them to:

1. Identify frequently queried columns.
2. Create appropriate indexes.
3. Explain clustered/non-clustered indexing where supported by their DBMS.
4. Compare query performance before and after indexing.
5. Use an execution plan.
6. Identify an inefficient query.
7. Optimize the query.
8. Explain the trade-off between indexing and write performance.

Deliverable: Before-and-after performance analysis.

Task 15: Database Governance Scenario

Governance problem:

A faculty member requests access to all student records because they are supervising several research projects. The faculty member argues that access to all student information will make research management easier.

Students must decide:

1. Should the request be approved?
2. What information should the faculty member receive?
3. What information should be restricted?
4. Who should approve the request?
5. What database roles should be used?

6. Should access be temporary?
7. Should access be audited?
8. What principle of database governance supports their decision?

This tests whether students can **apply governance principles rather than simply create SQL statements.**

Task 16: Create data/database ethics report (access, usage, storage, management, administration)

1. Phase 1: Database Implementation

ERD → relational schema → normalization → DDL → data → SQL.

2. Phase 2: Advanced Database Management

Views → procedures → functions → triggers → transactions → indexing/performance.

3. Phase 3: Database Security and Governance

Roles → privileges → least privilege → data classification → privacy → auditing → governance.

4. Phase 4: Administration and Final Project

Backup/recovery → disaster recovery → governance scenario → integrated demonstration.

5. Data/ database Ethical report